

Blood relations

COMPLETE LEARNING GUIDE

What You'll Learn

Blood relation questions test whether a candidate can build and traverse a family tree correctly across generations, often disguised through coded symbols (+, " , ×, ÷ standing for relations) or riddle-style phrasing ('pointing to a photograph...') designed to hide a simple relationship behind confusing wording. The core skill is mechanical, not clever — draw the family tree with a generation-line notation as each clue arrives, and the answer falls out by reading the tree rather than trying to hold the relationship chain in your head. This sub-module covers basic relation-term notation, multi-generation tree building, coded (symbolic) blood relations, the classic gender-ambiguity traps that catch students off guard, complex multi-generation puzzles, a quick-reference relation table, and the well-known 'pointing to a photograph' riddle format, each worked through to a definite answer.

1. Basic Relation Terms and Tree Notation

Build a simple tree using generation levels (older generation on top, younger below) connected by relation labels — this converts a string of stated relations into a visual structure that's easy to read the final answer from, rather than re-deriving the chain from scratch for each question about the same family.

- 'A is B's father. B is C's mother.' Draw A (top) connected down to B, and B connected down to C, with C's gender unspecified. A is C's maternal grandfather (father of C's mother).

2. Generation Tree Building Across Multiple Clues

As clues accumulate, keep adding to the same tree rather than starting a fresh mental calculation for each new statement — most errors happen when a student tries to re-derive earlier relationships from memory instead of reading them off an already-built diagram.

- 'P is Q's mother. R is P's father. S is R's wife.' Tree: R and S (grandparent generation) !' P (their child) !' Q (P's child). S, being R's wife, is P's mother, making S Q's grandmother (maternal, since P is Q's mother).

3. Coded Blood Relations (Symbol-Based)

Symbol-coded questions define operators (commonly +, -, ×, ÷) for specific relations at the start of the question — translate each symbol into its relation word first, then build the tree exactly as with plain-language clues.

- Given: A+B means A is B's mother; A-B means A is B's brother; A×B means A is B's father; A÷B means A is B's sister. Evaluate 'P × R + S': P×R means P is R's father. R+S means R is S's mother (so R is female). Combining: P is the father of R, and R is the mother of S — so P is S's maternal grandfather.

4. Gender Ambiguity and Classic Traps

Many blood relation riddles hinge on a subtle logical trick rather than a long chain — reading the statement literally and carefully, rather than assuming a 'typical' relationship, is essential.

- 'Pointing to a man, a woman said: "His mother is the only daughter of my mother."' The only daughter of the woman's mother is the woman herself (she has no sisters). So 'his mother' = the woman. The woman is the man's mother.

5. Complex Multi-Generation Puzzles

When a question spans three or more generations, build the tree top-down (oldest generation first) and resolve the final relationship by tracing the shortest path between the two people asked about, naming each intermediate relation as you go.

- 'A is the son of B. B is the sister of C. C is the mother of D.' B is C's sister, and C is D's mother, so B is D's (maternal) aunt. A is B's son, so A is D's cousin (specifically, the son of D's maternal aunt).

6. Quick-Reference Relation Chains

Certain compound relations recur constantly and are worth memorizing directly rather than re-deriving each time.

- Mother's/father's brother = uncle; mother's/father's sister = aunt; uncle's/aunt's child = cousin; spouse's brother = brother-in-law; spouse's sister = sister-in-law; father's father = paternal grandfather; mother's father = maternal grandfather.

7. The 'Pointing to a Photograph' Riddle Format

This classic format always describes a relationship of the speaker to a person in a photo, and the trick is almost always in correctly identifying who 'my grandfather's only son' or similar phrases actually refers to — usually the speaker's own father, unless the wording explicitly rules that out.

- 'Pointing to a photograph, a man said: "She is the daughter of my grandfather's only son."' The speaker's grandfather's only son is the speaker's own father (assuming the standard convention that the speaker isn't referring to himself). So 'she' is the speaker's father's daughter — his sister.

Common Pitfalls

- Trying to hold a multi-step relation chain in your head instead of drawing a generation tree as each clue arrives.
- Assuming a gender for an ambiguous relation term (e.g., 'sibling', 'cousin', 'child') that the question never actually specifies.
- Misreading 'only son'/'only daughter' clues, which are often the key trick identifying the speaker or a specific relative directly.
- In coded (symbolic) relation questions, forgetting to first translate every symbol into its relation word before building the tree.
- Losing track of which generation level a person belongs to in a 3+ generation puzzle, leading to an off-by-one-generation error.
- Assuming 'brother' or 'sister' when the question only established a parent-child or sibling-of-parent relationship without confirming gender.

Key Takeaways

- Always draw a generation-based family tree as clues arrive — never try to resolve blood relations purely mentally.
- In coded/symbolic questions, translate every symbol to its relation word first, then build the tree exactly as usual.
- 'Only son'/'only daughter' clues are almost always the key that identifies a specific person (often the speaker or a parent) precisely.
- Memorize the standard compound-relation chains (uncle, aunt, cousin, in-laws, grandparents) to skip re-deriving them each time.
- In photograph/riddle-style questions, default to the standard convention that the speaker is describing a relative other than themselves unless stated otherwise.
- Trace the shortest path between the two people asked about in a multi-generation tree, naming each link along the way.

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